classical mixing: the pile-up truth

signal GEN-SIM
Geant4 particles and hits

↓

minimum-bias GEN-SIM
the pile-up library of the campaign

↓

DIGI with the MixingModule
the campaign's pile-up profile, in-time and out-of-time crossings

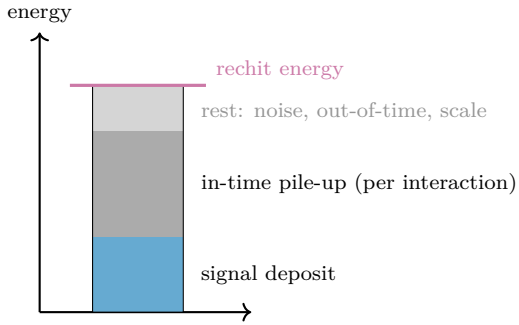
↓

picoMixedTruth, kept from the transient CrossingFrames:
in-time interactions: SimTracks, SimVertices, calorimeter sim hits, each tagged (interaction, Geant4 id)
out-of-time: deposits per (cell, crossing)
TrackingParticles of the pile-up too

↓

RECO → Pico
`pileupTruth=picoMixedTruth`

every calorimeter cell



the sim deposits are put on the rechit scale with factors measured here, cell by cell and per subdetector (the HF gain among them)

`SimCaloCell`: `energy`, `puEnergy`, `ootEnergy`, `nPU`

the jets with the pile-up in them (anti-k_T 0.4, matched to the gen jet through their signal constituents)

SimCaloCellPUJet: the cells with their signal + in-time pile-up deposits, the calorimeter truth with pile-up, no noise, no thresholds

SimPFPUJet: the ideal PF of the signal and the pile-up entrance particles (charged trackable from momentum, the rest from deposits), the counterpart of the raw PF jet

SimPFPUCHSJet: the same without the trackable charged pile-up, a perfect charged-hadron subtraction, the counterpart of the CHS jet: what remains is the neutral and untrackable pile-up

the pile-up of one jet, three ways

1. attribution (any sample)
`ptRatio` − `signalPtRatio`: the raw jet minus its signal part, candidate by candidate

2. truth, no reconstruction (mixed sample)
$R_{\text{SimPFPU}} - R_{\text{SimPF}}$: the ideal PF with and without the pile-up particles;
$R_{\text{SimPFPUCHS}}$: after a perfect charged-hadron subtraction

3. truth through the reconstruction (mixed sample)
`puFraction` of the constituents: the offset split into true in-time pile-up by candidate type and a residual (noise, out-of-time, scale)

2 − 1: what the reconstruction adds to the pile-up (thresholds, clustering, calibration)
1 − 3: the attribution convention
SimPFPU vs PF and SimPFPUCHS vs CHS: the reconstruction and the vertex assignment at work; Puppi has no ideal of its own and is read against SimPF